

TCS349: Responsible and Explainable AI

Unit 2, Lectures 4 & 5

Accountability in AI: Roles, Oversight & Governance

Who is responsible when an AI system causes harm - and how organisations build the frameworks, oversight and compliance structures that make accountability real.

Mr. Manoj Kaushik
Assistant Professor, Department of CSE
Graphic Era Hill University, Bhimtal



Roles & Responsibility



Human Oversight



Escalation



Governance & Compliance

2. What Is Accountability in AI?

Accountability is the obligation to answer for an AI system's decisions, actions and outcomes, including any harm it causes.

It requires that responsibility be assigned to specific people or roles, not left to the system itself.

An accountable organisation can always answer: who approved this system, who monitors it, and who acts when something goes wrong.

Accountability turns responsible-AI principles into an enforceable obligation rather than a voluntary aspiration.

3. Roles & Responsibility Across the AI Lifecycle

Data team: accountable for the quality, consent and representativeness of the data used.

ML/engineering team: accountable for model design, testing and documented performance, including fairness checks.

Product owner: accountable for defining intended use and approving deployment.

Deploying organisation: accountable for ongoing monitoring, user communication and responding to harm after launch.

4. Human Oversight - Why It's Needed

Human oversight means a person retains the ability to review, question or override an AI system's decisions.

It acts as a safeguard against errors, edge cases and biases that automated testing did not catch.

Oversight is especially critical in high-stakes domains such as healthcare, lending, hiring and criminal justice.

Without oversight, accountability has no practical mechanism to act on.

5. Levels of Human Oversight

Level	What It Means
Human-in-the-Loop	A human reviews and approves every individual decision before it takes effect
Human-on-the-Loop	A human monitors the system in real time and can intervene, but does not approve every decision
Human-in-Command	A human retains overall control and can override, disable or halt the system at any point

6. Escalation Mechanisms - What They Are

An escalation mechanism is a defined path for routing a decision or concern from the automated system to a responsible human.

It specifies who is notified, under what conditions, and how quickly a response is required.

Escalation is what turns human oversight from a general capability into a specific, working process.

A well-designed escalation path exists before a system is deployed, not after a failure occurs.

7. Designing an Escalation Path

Trigger: a decision the model is highly uncertain about, or one it rejects, above a defined threshold.

Route: the case is flagged and sent to a named reviewer or review queue, not left unresolved.

Response time: a maximum time limit is set for the human reviewer to act.

Record: the escalation, reviewer decision and reasoning are logged in the system's audit trail.

8. Summary: Accountability Basics

Accountability assigns clear responsibility for an AI system's decisions across defined roles.

Human oversight, at an appropriate level, is the safeguard that accountability depends on.

Escalation mechanisms turn oversight into a specific, working process rather than a vague promise.

Next: how organisations formalise accountability through frameworks, governance and compliance.

9. From Accountability to Accountability Frameworks

Individual roles and escalation paths work at the level of one system - frameworks operate at the level of an entire organisation.

An accountability framework standardises how every AI system in an organisation is reviewed, approved and monitored.

It prevents accountability from depending on the judgement of one team or individual alone.

This is the bridge from Lecture 13's practices to Lecture 14's structures.

10. What Is an Accountability Framework?

An accountability framework is a formal structure defining roles, review processes and documentation required before, during and after deployment.

It typically includes a review board or committee, a risk-tiering process, and required sign-offs at key stages.

Higher-risk systems - such as those affecting employment, credit or safety - usually require stricter review under the framework.

Well-known reference points include the NIST AI Risk Management Framework and organisation-specific responsible-AI policies.

11. Organisational Governance Structures

Responsible AI review boards: cross-functional groups that evaluate a system's risk before approving deployment.

Internal policies and standards: written rules defining minimum requirements for fairness, transparency and documentation.

Designated ownership: a named individual or team accountable for each deployed system, not a diffuse group.

Training and awareness: ensuring staff across the organisation understand their role in responsible AI.

12. Regulatory Compliance - Why It Matters

Regulatory compliance is the minimum legal baseline an AI system must meet - it does not replace organisational governance, which often goes further.

Regulations increasingly require documented risk assessments, human oversight and the ability to explain automated decisions.

Non-compliance can carry legal penalties, but also reputational and operational harm.

Compliance requirements differ by sector and region, so accountability frameworks must be adaptable.

13. Governance and Compliance Working Together

Layer	Example
Regulation	Data-protection law requiring notice before an automated decision affecting a person
Organisational Policy	Internal rule requiring subgroup fairness testing before any model is approved
Team Practice	ML team documents fairness results in a model card before requesting sign-off
Escalation Record	Reviewer logs the reason for overriding a specific automated decision